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2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

Task 1:

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ nano practical2.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ gcc practical2.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ ./a.out
File copied successfully
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ |
```

CONTENTS IN SOURCE.TXT(b2):

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ cat b2
This is the program related to System calls
It is about open read and write System calls
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ |
```

CONTENTS IN DESTINATION.TXT(c2):

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ cat c2
This is the program related to System calls
It is about open read and write System calls
```

CODE:

```
GNU nano 8.7.1
#include<stdio.h>
#include<fcntl.h>
#include<unistd.h>
int main()
{
int fd1, fd2;
size_t n;
char buf[100];
fd1 = open("b2", O_RDONLY);
fd2 = open("c2", O_WRONLY | O_CREAT, 0644);
while((n = read(fd1, buf, sizeof(buf))) > 0)
{
write(fd2, buf, n);
}
close(fd1);
close(fd2);
printf("File copied successfully\n");
return 0;
}
```

OUTPUT:

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ nano practical2.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ gcc practical2.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ ./a.out
File copied successfully
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ cat b2
This is the program related to System calls
It is about open read and write System calls
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ cat c2
This is the program related to System calls
It is about open read and write System calls
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ |
as is about open read and write system calls
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ echo "Hello World" > sample.txt
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ cat sample.txt
Hello World
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ strace cat sample.txt
Command 'strace' not found, but can be installed with:
sudo apt install strace
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ |
```

Execution : Task 2

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffd3a6d2038 /* 25 vars */) = 0
brk(NULL) = 0x64c0dac78000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7796633b4000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20085, ...}) = 0
mmap(NULL, 20085, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7796633af000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x77966337c000
mmap(0x779663383000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x779663383000
mmap(0x7796633a4000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7796633a4000
mmap(0x7796633ab000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x7796633ab000
mmap(0x7796633ad000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7796633ad000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x77966334e000
mmap(0x779663352000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x779663352000
mmap(0x779663376000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x779663376000
mmap(0x77966337a000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x77966337a000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x779663228000
mmap(0x77966323a000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x77966323a000
mmap(0x7796632da000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x7796632da000
mmap(0x77966334c000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x77966334c000
```

```
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x779663000000
mmap(0x779663028000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x779663028000
mmap(0x7796631c0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x7796631c0000
mmap(0x77966320e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x77966320e000
mmap(0x779663214000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x779663214000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x779662f54000
mmap(0x779662f57000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x779662f57000
mmap(0x779662fd3000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f000) = 0x779662fd3000
mmap(0x779662ffe000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x779662ffe000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x779663226000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x779663223000
arch_prctl(ARCH_SET_FS, 0x779663223b00) = 0
set_tid_address(0x779663224128) = 691
set_robust_list(0x779663223de0, 24) = 0
rseq(0x7796632236a0, 0x20, 0, 0x53053053) = 0
mprotect(0x77966320e000, 16384, PROT_READ) = 0
mprotect(0x779662ffe000, 4096, PROT_READ) = 0
mprotect(0x77966334c000, 4096, PROT_READ) = 0
mprotect(0x77966337a000, 4096, PROT_READ) = 0
mprotect(0x7796633ab000, 4096, PROT_READ) = 0
mprotect(0x64c09accc000, 1462272, PROT_READ) = 0
mprotect(0x7796633f9000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
```



```

getrandom("\x33\xe0\x01\x94\x12\x3c\x43\x5a", 8, GRND_NONBLOCK) = 8
munmap(0x7796633af000, 20085) = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0x37351577, 0x142ff173]}, f_namele=255, f_frsize=4096, f_flags=ST_VALID|ST_NOSUID|ST_NODEV|ST_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7fffa382b900) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x64c0dac78000
brk(0x64c0dac99000) = 0x64c0dac99000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"... , 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-"... , 1024) = 1024
read(3, "ime 0 0\ndevpts /dev/pts devpts r"... , 1024) = 1024
read(3, "xec,relatime 0 0\nhugetlbfs /dev/"... , 1024) = 1024
read(3, "mpfs /mnt/wslg/run/user/1000 tmp"... , 1024) = 122
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
poll([fd=0, events=0], {fd=1, events=0}, {fd=2, events=0}, 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x779663045cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "64c09a363000-64c09a455000 r--p 0"... , 1024) = 1024

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read(3, "2ffe000-779662fff000 r--p 000a90"... , 1024) = 1024
read(3, "00 08:30 14388"... , 1024) = 1024
read(3, " /usr/lib/x86_64-linux"... , 1024) = 1024
read(3, "0 13931 /us"... , 1024) = 566
close(3) = 0
sched_getaffinity(691, 32, [0 1 2 3 4 5 6 7]) = 8
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7796633b1000
mprotect(0x7796633b1000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x7796633b2000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid() = 691
rt_sigaction(SIGSEGV, {sa_handler=0x64c09a6880c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x779663045cb0}, NULL, 0) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=0x64c09a6880c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x779663045cb0}, NULL, 8) = 0
prctl(PR_GET_AUXV, 0x7fffa382b3f0, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attribute=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/uucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attribute=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x779663045cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x779663045cb0}, 8) = 0

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close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attribute
s=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x779663045cb0}, {sa_handler=
SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x779663045cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x779663045cb0}, {sa_handler=0x64
c09a6880c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x779663045cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x779663045cb0}, {sa_handler=0x64c
09a6880c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x779663045cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|EC
HO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x64c0dacba000) = 0x64c0dacba000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|
0644, stx_size=12, ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=12, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 12
splice(4, NULL, 1, NULL, 12, 0Hello World
) = 12
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7796633b1000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ |

```